Cycle deformation, monodromy detection, and exact bidisk realization

24 September 2026. This note supplies a global transition test for the allowed product branch. Small deformation on individual balls is replaced by a bound accumulated around actual generators of the base fundamental group. The test detects finite mapping-class monodromy; it does not assume it, assume a nearby Einstein metric, or require solving the tensor equation near the input metric. Classical surface theory then realizes an exact Einstein metric on the original smooth total space.

The retained Round 18 theorem starts with finite monodromy and describes the complete Einstein operator and nearby solutions. Its conclusion is not being rediscovered here. The additional bridge is from a differential cycle budget to that global topological hypothesis. Two uses of the inherited Dehn-twist construction show independently why local smallness and degeneration of the fiber cannot be ignored. No claim about the nonexistence of general Einstein metrics on those examples is made.

1. An intrinsic cycle budget

Let

$$F_f \longrightarrow M \xrightarrow{\pi} B_b \qquad (f, b \geqslant 2) \tag{1}$$

be a smooth oriented bundle of closed connected oriented surfaces. Let g be any smooth metric on M, let $V = \ker d\pi$, and use the orthogonal Ehresmann connection $H = V^{\perp}$. A Riemannian submersion, horizontal integrability, or constant fiber curvature is not required in the main theorem. Compactness makes horizontal lifting complete along each finite path.

For a piecewise smooth based loop $\gamma : [0,1] \to B$, let $\Phi_t : F_0 \to F_{\gamma(t)}$ be horizontal transport, with $F_0 = F_{\gamma(0)}$. Define metrics on the fixed surface F_0 by

$$h_t = \Phi_t^*(g|_{F_{\gamma(t)}}), \qquad A_t = h_t^{-1}\dot{h}_t, \qquad A_t^0 = A_t - \frac{1}{2}\operatorname{tr}(A_t)I,$$

and define the conformal cycle variation

$$\mathscr{J}_g(\gamma) = \frac{1}{2}\int_0^1 \sup_{y \in F_0} \|A_t^0(y)\|_{\mathrm{op}}\, dt. \tag{2}$$

The expression is invariant under orientation-preserving reparametrization of the loop and under a fixed change of marking of F_0. It measures variation of its conformal structure, not a change in total fiber area. No choice of a representative of a formal monodromy class replaces the actual transport in (2).

Let $\Gamma = (\gamma_1, \dots, \gamma_N)$ be based loops at a **common** point b_0, whose homotopy classes generate $\pi_1(B, b_0)$. Put

$$\mathscr{J}_g(\Gamma) = \max_i \mathscr{J}_g(\gamma_i). \tag{3}$$

Let $x_0 \in \mathscr{T}_f$ be the marked conformal class of $g|_{F_0}$. The thickness condition below refers to the systole of the unique curvature -1 metric in this conformal class. It is not a lower bound for four-dimensional balls or an injectivity-radius bound for the given metric.

Theorem A (global monodromy detection and realization). For every $f \geqslant 2$ and $\iota > 0$, there is $\varepsilon_{f,\iota} > 0$ such that the following holds. If (1) has a metric and generating loops with

$$\operatorname{sys}(x_0) \geqslant \iota, \qquad \mathscr{J}_g(\Gamma) < \varepsilon_{f,\iota}, \tag{4}$$

then its mapping-class monodromy has finite image G. The original smooth bundle is smoothly isomorphic to

$$(\widehat{B} \times F_f)/G \longrightarrow B, \tag{5}$$

where $\widehat{B} \to B$ is the connected regular cover associated to the monodromy kernel, G acts freely on $\widehat{B}$, and its action on F_f is by orientation-preserving hyperbolic isometries. Consequently the same smooth M carries an exact metric h with

$$\mathrm{Ric}_h = -h, \qquad (\widetilde{M}, \tilde{h}) \cong \mathbb{H}^2_{-1} \times \mathbb{H}^2_{-1}. \tag{6}$$

It is Kähler for two opposite orientations. No closeness between g and h is asserted. The threshold is independent of the genus of the base, the number of generators, diameter, total volume, and the order of the eventually detected finite group.

In particular, a pointwise near-Einstein input satisfying (4) has the Einstein output required by Yau26 in this input class. In fact (4) alone is sufficient: the theorem does not need the near-Einstein hypothesis. Deriving (4) from a weaker original geometric input is a separate task; Sections 4–6 identify what data it must retain. The companion Ricci-to-monodromy theorem performs this step for minimal surface submersions, using the horizontal Ricci identity and a based short-generator argument. Its hypothesis is $2\,\mathrm{diam}(B)\sqrt{\delta + \|(K_B+1)_+\|_\infty} < \varepsilon_{f,\iota}$, with the same constant as below.

2. The exact Teichmüller displacement estimate

We use $d_T(x,y) = \frac{1}{2}\inf \log K(\phi)$, where the infimum is over quasiconformal maps in the prescribed marking class. Write $\rho : \pi_1(B) \to \mathrm{Mod}(F_f)$ for the monodromy.

Lemma 1. For every based loop,

$$d_T(x_0, \rho(\gamma)x_0) \leqslant \mathscr{J}_g(\gamma). \tag{7}$$

The convention for the mapping-class action may replace $\rho(\gamma)$ by its inverse; the displacement is unchanged.

Proof. For any two nonzero fixed tangent vectors v, w at a point of F_0, differentiating the ratio of their squared lengths gives

$$\left| \frac{d}{dt} \log \frac{h_t(v,v)}{h_t(w,w)} \right| \leqslant 2\|A^0_t\|_{\mathrm{op}}. \tag{8}$$

At a fixed point take vectors attaining the largest and smallest length ratios between h_1 and h_0. Integration of (8) implies that the logarithm of the quasiconformal dilatation of the identity between these two conformal metrics is at most $\int \sup \|A^0_t\|_{\mathrm{op}} dt$. The metric h_1 is the pullback of $g|_{F_0}$ by the actual return map Φ_1, whose mapping class is $\rho(\gamma)$. The definition of d_T therefore gives (7). This proof does not discard rotations of the principal directions or require that the matrices A_t commute. □

Although the connection need not be flat, isotopy classes of its return maps are invariant under based homotopy of loops and obey the group relations. Indeed horizontal lifting over a smooth homotopy gives an isotopy of the resulting fiber diffeomorphisms. Thus the actual classes used in (7) form the monodromy representation, not an unrelated list of almost-isometries.

3. Small moves in a thick fiber force one finite group

Lemma 2 (uniform finite-isotropy neighbourhood). For fixed f, ι, some $\varepsilon_{f,\iota} > 0$ has the following property. If $x \in \mathscr{T}_f$, $\mathrm{sys}(x) \geqslant \iota$, and every member of a set $S \subset \mathrm{Mod}(F_f)$ moves x by less than $\varepsilon_{f,\iota}$, then $\langle S \rangle$ is finite. The number of elements of S need not be bounded.

Proof. We use two classical inputs: the mapping-class action on Teichmüller space is proper, with finite stabilizers, and the quotient of the fixed-systole thick part is compact by Mumford's compactness theorem. Thus a compact set $K \subset \mathscr{T}_f$ meets every thick orbit.

If the lemma failed, after simultaneous conjugation there would be $x_j \in K$, $x_j \to x$, and sets S_j generating infinite groups, with $d_T(x_j, sx_j) < 1/j$ for all $s \in S_j$. For these elements,

$$d_T(x, sx) \leqslant 2d_T(x, x_j) + 1/j \longrightarrow 0 \tag{9}$$

uniformly in s. Properness gives a neighbourhood U of x such that only finitely many group elements can carry a smaller compact neighbourhood into one meeting U. Equivalently, since d_T is proper, the set $\{s : d_T(x, sx) \leqslant 1\}$ is finite. Its nonzero displacements have a positive minimum. Hence, for large j, every member of S_j lies in $\mathrm{Stab}(x)$. This finite group contains $\langle S_j \rangle$, a contradiction. A possible finite kernel of the action, including the genus-two hyperelliptic involution, causes no problem. □

Proof of Theorem A. Lemmas 1–2 place all the generating monodromies in a single finite group. It is insufficient to check that each generator has finite order separately; Lemma 2 is precisely what controls the group they generate.

Kerckhoff's Nielsen realization theorem realizes the finite image G as isometries of one curvature -1 metric on the fiber. Give B any hyperbolic metric and $\widehat{B}$ its lift. The diagonal action on $\widehat{B} \times F_f$ is free because its first component is a deck action, even if its fiber action has fixed points. The product metric descends smoothly and satisfies (6).

It remains to identify the smooth total space, not merely its group. The identity component $\mathrm{Diff}_0^+(F_f)$ is contractible by Earle–Eells. Therefore $B\,\mathrm{Diff}^+(F_f) \simeq B\,\mathrm{Mod}(F_f)$, and oriented smooth surface bundles over B are classified by conjugacy classes of their monodromy representations. The original bundle and the constructed suspension have the same representation, hence are smoothly bundle-isomorphic. Pulling back the descended metric proves existence on the original M. The two product complex structures are invariant under the orientation-preserving diagonal action, so both descend. □

This proof is a composition with classical global surface theory, not a new Nielsen realization theorem. It uses neither a global tensor inverse nor the assumption that a candidate Einstein metric preserves the input bundle. It constructs one that does; that suffices for existence.

4. Differential and opposite-form versions

Let $\mathrm{II}(Y, Z) = (\nabla_Y Z)^H$ for vertical Y, Z. For a horizontal vector X, define the symmetric vertical endomorphism S_X by

$$g(S_X Y, Z) = g(\mathrm{II}(Y, Z), X).$$

For the projectable horizontal field that lifts $\dot{\gamma}$,

$$(\mathscr{L}_X g)|_{V \times V} = -2g(S_X \cdot, \cdot), \quad \mathscr{J}_g(\gamma) = \int_0^1 \sup_{F_{\gamma(t)}} \|S_X^0\|_{\mathrm{op}} dt. \tag{10}$$

Indeed metric compatibility gives $g(\nabla_Y X, Z) = -g(X, \nabla_Y Z)$; the symmetric sum is the Lie derivative. Pulling this identity back by Φ_t gives (10).

Consequently, for a Riemannian submersion onto (B, k),

$$\mathscr{J}_g(\Gamma) \leqslant \Lambda_k(\Gamma) \sup_{|X|=1,\ X\in H} \|S^0_X\|_{\mathrm{op}}, \qquad \Lambda_k(\Gamma) = \max_i \mathrm{Length}_k(\gamma_i). \tag{11}$$

Both factors matter. One cannot replace the right side by the local deformation amplitude alone on a base with arbitrarily long essential loops.

Here is a form-level version that keeps exact integrability explicit. Suppose two smooth forms satisfy

$$*\omega_+ = \omega_+, \quad *\omega_- = -\omega_-, \quad |\omega_\pm|^2 = 2, \quad d\omega_\pm = 0. \tag{12}$$

Put $\alpha = (\omega_+ + \omega_-)/2$ and $\beta = (\omega_+ - \omega_-)/2$. Then α, β have rank two, $\alpha^2 = \beta^2 = 0$, and $\alpha \wedge \beta$ never vanishes, up to the choice of orientation/sign. Their kernel distributions are transverse, orthogonal, and integrable: if a closed constant-rank two-form annihilates Y, Z, Cartan's formula gives $\iota_{[Y,Z]}\alpha = 0$. This is an exact symplectic pair, not an inference from approximate closedness.

Assume one distribution is the vertical tangent bundle of a smooth surface bundle as in (1). This compact-leaf/leaf-space condition is not automatic. If P is its orthogonal projection and $J_\pm$ are the endomorphisms associated with $\omega_\pm$, then, with one of the two signs,

$$P = \tfrac{1}{2}(I \pm J_+ J_-), \qquad |\nabla P| \leqslant \frac{1}{\sqrt{2}}(|\nabla \omega_+| + |\nabla \omega_-|). \tag{13}$$

We use Hilbert–Schmidt norms for tensors and the standard exterior norm for forms; $|\nabla J|^2 = 2|\nabla \omega|^2$. Also $\|S^0_X\|_{\mathrm{op}} \leqslant |X||\nabla P|$. Thus, under the Riemannian-submersion hypothesis, Theorem A applies if

$$\frac{\Lambda_k(\Gamma)}{\sqrt{2}}\left(\|\nabla \omega_+\|_\infty + \|\nabla \omega_-\|_\infty\right) < \varepsilon_{f,\iota}. \tag{14}$$

The sharper loopwise test (10) is preferable when deformation is inhomogeneous. No bound on second derivatives of these forms is used.

The actual-form supply theorem currently gives mean derivative bounds; the collapsed-bulk theorem gives local limiting parallel forms. Neither alone proves (12), the existence of the global fibration, or the cycle estimate (10). The present theorem specifies a genuine sufficient global input rather than declaring those missing implications automatic.

5. Sharpness I: thick fibers and uniform near-parallelism do not suffice

We extract additional consequences of the inherited fixed bundle E_D, not a newly constructed smooth object. Fix $f, b \geqslant 2$, a positive Dehn twist D about a nonseparating curve of F_f, and the epimorphism $u : \pi_1(B_b) \to \mathbb{Z}$ that sends one standard generator to 1 and all others to 0. Its monodromy is $[D]^{u(\gamma)}$. Choose a basepoint of the fiber off the twist support, so the suspension has a section and

$$\pi_1(E_D) = \pi_1(F_f) \rtimes \pi_1(B_b), \qquad b_1(E_D) = 2f + 2b - 1. \tag{15}$$

The last equality follows because a nonseparating twist acts by a rank-one transvection on $H_1(F_f; \mathbb{R})$; its coinvariant space has dimension $2f - 1$. In particular E_D admits no Kähler metric, since the first Betti number of a compact Kähler manifold is even. This simple obstruction is sufficient here; the stronger inherited finite-cover obstructions are not needed.

Use the inherited area-preserving hyperbolic twist path on a fixed fiber:

$$h_z = du^2 + a(u)^2(ds + z\zeta'(u)du)^2, \quad a(u) = a_0 \cosh u, \quad h_1 = D^* h_0, \quad dA_{h_z} = a(u)du \wedge ds. \tag{16}$$

It is extended by the fixed hyperbolic metric outside the collar. The path $z \in [0,1]$ is compact, so all its hyperbolic systoles have a common positive lower bound. These are not globally trivial deformations: an intermediate collar rotation is not a global isotopy that removes D.

On a long pinching base collar set

$$g_L = dt^2 + b_L(t)^2 d\theta^2 + h_{z(t)}, \qquad b_L(t) = \ell \cosh t, \quad z(t) = \eta(t/L), \tag{17}$$

where η is monotone from 0 to 1 and constant near the ends. The source construction closes (17) by the marked Dehn twist, on the fixed E_D, and proves

$$\| \operatorname{Ric}^{\sharp}_{g_L} + I \|_\infty = O(L^{-1}), \qquad \operatorname{Vol}(g_L) = 16\pi^2 (f-1)(b-1). \tag{18}$$

No region-volume argument is used for this maximum-norm estimate.

The horizontal and vertical area forms α_L, β glue globally, are closed, and satisfy

$$\omega_{+,L} = \alpha_L + \beta, \qquad \omega_{-,L} = \alpha_L - \beta, \quad *\omega_{\pm,L} = \pm\omega_{\pm,L}, \quad |\omega_{\pm,L}|^2 = 2. \tag{19}$$

Closedness uses the **fixed pointwise fiber area form** in (16), not just its fixed integral. The marked gluing preserves that form. The horizontal distribution is flat, and the only mixed second fundamental terms are produced by $\partial_t h_{z(t)}$. Writing $Q = a\zeta'$, $v = Qz'$, their orthonormal matrix is

$$S_{\partial_t} = -\frac{v}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{20}$$

Thus $|\nabla P| = |v|$, and the connection on the two rank-two blocks gives $|\nabla \omega_{\pm,L}| \leqslant C|v|$. Therefore

$$\|\nabla \omega_{\pm,L}\|_\infty = O(L^{-1}). \tag{21}$$

The forms are also harmonic, by (19) and closedness.

Nevertheless (15) prohibits any exact Kähler output. Along a loop that crosses the twist collar once and otherwise remains in the product region, (20) and monotonicity give the **nonvanishing** cycle budget

$$\mathscr{I}_{g_L}(\gamma) = \frac{1}{2} \|Q\|_\infty. \tag{22}$$

In particular it is at least $d_T(x_0, Dx_0) > 0$. An infinite-order twist cannot fix a hyperbolic conformal structure, whose automorphism group is finite. The deformation amplitude in (21) vanishes, but the length over which it accumulates diverges. At least one essential cycle retains the global information.

Thus even **closed, nowhere-zero, exactly normalized opposite forms**, uniform near-parallelism, pointwise near-Einstein curvature, and thick fiber conformal classes do not imply global KE realization without control of transition cycles. This is stronger than merely observing that an exceptional set in an average theorem might carry topology.

6. Sharpness II: vanishing cycle distortion does not replace fiber thickness

Keep the same smooth bundle and now keep a fixed hyperbolic base and a fixed base collar. Pinch the twisting curve in the fiber to length $\lambda \downarrow 0$. In its hyperbolic collar use the angular period 1 and

write $a_\lambda(u) = \lambda \cosh u$. Fix a nonnegative smooth χ, supported in $(-u_0, u_0)$, with positive integral, and choose

$$C_\chi = \int_{-u_0}^{u_0} \chi(u) \operatorname{sech} u \, du, \qquad \zeta'(u) = \frac{\chi(u)}{C_\chi \cosh u}. \tag{23}$$

The integral of ζ' is 1, so the return map is still one full Dehn twist, not a small fractional topological twist. But

$$Q_\lambda = a_\lambda \zeta' = \lambda \chi / C_\chi \tag{24}$$

and all its fixed-order u-derivatives are $O(\lambda)$. The increasingly wide hyperbolic collar contains this fixed support. Extend the hyperbolic fiber metrics using any fixed marked pinching family; the construction remains on the same smooth bundle because its mapping class remains $[D]$.

Take a fixed smooth monotone $z(t)$ across the fixed base collar. For completeness, in the orthonormal frame of (17) the inherited exact Ricci defect is

$$\operatorname{Ric} + g = \begin{pmatrix} -v^2/2 & 0 & 0 & (v_u + 2jv)/2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -v^2/2 & -(v_t + kv)/2 \\ (v_u + 2jv)/2 & 0 & -(v_t + kv)/2 & v^2/2 \end{pmatrix}, \quad j = \tanh u, \ k = b'/b, \ v = Q_\lambda z'. \tag{25}$$

All the functions on the fixed base collar have bounded derivatives. Equations (24)–(25), together with the product metric off the supports, therefore give

$$\delta_-(g_\lambda) = O(\lambda), \qquad \|\nabla \omega_{\pm,\lambda}\|_\infty = O(\lambda), \qquad \mathscr{J}_{g_\lambda}(\Gamma) = O(\lambda). \tag{26}$$

A fixed standard based generating system has bounded base lengths. Its only nontrivial return map is D; the other loops can be chosen to avoid the active collar or to run along it at constant t. In (26) the forms still satisfy all the exact conditions (19).

The image of monodromy is nevertheless infinite cyclic, and (15) still rules out Kähler geometry. The missing hypothesis is precisely the fiber conformal thickness: its systole is at most λ. This is an explicit small-displacement failure at the boundary of moduli space, not a failure of Lemma 2. Both circle collapse and long fiber diameter are genuine; the base geometry does not degenerate.

These are also actual bilateral near-saturation examples

Both constructions have symplectic structures (19) in the two orientations. They have

$$\tau(E_D) = 0, \quad A = 2\chi(E_D) = 8(f-1)(b-1) > 0, \quad V = 2\pi^2 A, \quad p_\pm \geqslant A. \tag{27}$$

For the signature, the chosen monodromy extends over a handlebody because it factors through one free generator; the surface bundle is the boundary of the resulting oriented five-dimensional bundle. For the class supply, $b_+ = b_- > 1$, so Taubes's symplectic canonical classes have nonzero ordinary SW invariants in the two orientations. They are actual for every metric and throughout the conformal weight construction used in the inherited first-line estimates. Their squares are A, hence their positive projected squares are at least A. These class-supply facts and the weighted-solvability passage are the same inherited classical inputs, not a new gauge theorem here.

Write $q_\pm$ for the bilateral normalized deficits of I25/I30. The exact inherited energy identity gives

$$q_\pm = \frac{9}{2V}\left[\int |\operatorname{Ric}|^2 - 8\pi^2(2p_\pm - A)\right].$$

Since $\fint |\mathrm{Ric}|^2 \leqslant 4(1+\delta_-)^2$,

$$0 \leqslant q_\pm \leqslant 36\delta_- + 18\delta_-^2 \to 0. \tag{28}$$

Thus the two sharpness examples lie in the actual bilateral gauge near-saturation regime, not only in an abstract almost-product class. They do **not** disprove Yau26: arbitrary non-Kähler Einstein existence on E_D remains undecided by these statements.

7. What this bridge changes

The inputs (4) are nonempty and not merely the equation for an existing Einstein metric. Every finite-monodromy bundle has a connection and invariant fiber metric with $\mathscr{J} = 0$; the base metric and its diameter can be arbitrary. Smooth changes of the metric and connection retain (4) for a fixed generating system when sufficiently small, while the input need not be Einstein. More generally, trace changes of fiber metrics do not affect (2), so input scalar curvature and scale are not required to be close to a product normalization.

Unlike Round 18, one need not know the monodromy image in advance, supply an exact reference metric, or discard slow Teichmüller coefficient functions. Their cumulative deformation is precisely what (2) measures. The proof constructs the global quotient only after this data has forced finite monodromy. It does not classify all nearby Einstein metrics or provide a uniform inverse on the quotient; those remain distinct claims in Round 18.

For the first line, the resulting mathematical choice is specific: local product estimates must be accompanied by a mechanism controlling global cycle distortion and escape toward the boundary of fiber moduli. Making every local form derivative small, even uniformly with exact closedness, does not remove those two independent alternatives. The actual examples above rule out that proposed shortcut while Theorem A gives a sufficient global replacement. No new smooth topology or world-first priority is asserted for this composition.

8. Sources and attribution

- D. Mumford, *A remark on Mahler's compactness theorem*, Proceedings of the AMS **28** (1971), 289–294. The fixed-genus hyperbolic systole criterion supplies compact thick moduli. Primary article.
- S. P. Kerckhoff, *The Nielsen realization problem*, Annals of Mathematics **117** (1983), 235–265, Theorems 4–5. The original text also records proper discontinuity and finite isotropy of the mapping-class action. Primary article; full text.
- C. J. Earle and J. Eells, *The diffeomorphism group of a compact Riemann surface*, Bulletin of the AMS **73** (1967), 557–559, Theorem 1(c); and *A fibre bundle description of Teichmüller theory*, Journal of Differential Geometry **3** (1969), 19–43. The contractible identity component supplies the smooth bundle classification used above. 1967 primary announcement; 1969 article.
- The actual fixed E_D construction, the full Ricci matrix, and the topology/gauge inputs are retained in the global paper, Section 7, and the detailed construction, labels `BUN:slow`, `BUN:ricci-matrix`, `BUNTOP:object`. The two new cycle-budget comparisons are consequences of those inputs.
- Bilateral compact realization and collapsed bilateral bulk have different hypotheses and conclusions from this note. Neither is silently applied to infer a smooth global leaf space.

The accompanying finite check verifies the two-dimensional variation normalizations, the shear

shape matrix, the transvection rank, and (28). It does not verify bundle classification, gauge solvability, or the analytic existence argument by computation.